Amazon as a Monopsony: Exhibiting Market Power over 3P Sellers

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What is a Monopsony?

- A Monopsony is the opposite of a Monopoly. Instead of a single seller, you have a single buyer dominating a market.

- When you only have one firm or individual to sell to, whether good, service, or human capital, that affords them a great deal of market power.

- A key point is that a monopsony can refer to people in ways a monopoly typically does not. One could observe a monopsonist as the sole employer of a particular advanced skill set in a given area. E.g. astronaut, quantitative analyst, etc. If there is only one large bank or space program in your area, they likely have a great deal of control over the price of your wages and intellectual capital.

Amazon as a Reimagined Monopsony pt.1

- To analyze Amazon, we must transition from a consumer-centric antitrust lens to a **Platform Monopsony** framework. Traditional metrics like HHI and consumer price indices are insufficient for digital markets where power is exerted on the supply side. Unlike traditional monopsonists that suppress input prices for labor or goods, Amazon exerts **Gatekeeper Power** effectively controlling the infrastructure of exposure. By selling access to a consolidated consumer base rather than purchasing seller inventory, Amazon creates a dependency that allows for the systematic extraction of economic rents through non-negotiable fee structures.

Amazon as a Reimagined Monopsony pt.2

Feature	Traditional Monopsony	Amazon (Platform Monopsony)
Source of Dominance	Buying volume	Access to customers (Demand)
Mechanism	Lowering input prices	Increasing platform fees
Key Metric	Market share of buyers	Share of "Search Real Estate"
Harm	Suppliers produce less	Sellers' margins are extracted
Victim	Worker/Supplier	3P Seller
Primary Constraint	Lack of local alternatives	Lack of access to buyers

The Antitrust Paradox of Ecommerce Platforms

- Amazon has often escaped Monopoly scrutiny from various Antitrust Laws because consumer prices remain low. While that is true, I think that is largely possible because of sellers experiencing high “extraction”.

- Consequently, it may more prudent to look at **Seller Dependency Ratios** to investigate Amazon’s market power rather than metrics like HHI (Herfindahl-Hirschman Index) that are based on consumer revenue.

Research Question

To what extent does Amazon’s market share in e-commerce retail allow it to exert monopsony power over third-party sellers?

The economic intuition is that Amazon's market share in the ecommerce market affords them a significant degree of power which they exercise to charge sellers, many of which having limited options outside of Amazon, fees above the market rate.

Regression

These coefficients may be interpreted as the % increase in take rate given a 1% increase in market share and/or logistics intensity.

Category Fixed Effects Panel Regression:

$Y_1(\text{Take_Rate}) = \alpha_1 + B_1\text{Market_Share} + B_2\text{Logistics_Intensity} \mid \text{Category} + \varepsilon_1$

```
OLS estimation, Dep. Var.: total_take_rate
Observations: 100
Fixed-effects: category: 10
Standard-errors: IID
                      Estimate Std. Error t value   Pr(>|t|)
market_share          0.335134   0.049787 6.73139 1.6478e-09 ***
logistics_intensity_gmv 0.002175   0.000517 4.21097 6.1227e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.022311     Adj. R2: 0.66585
                 Within R2: 0.555514
```

Categories: Apparel, Footwear & Accessories; Auto & Parts; Books, Music & Video; Computer & Consumer Electronics; Food & Beverage; Furniture & Home Furnishings; Household & Personable Consumables; Home Improvement & Garden; Office Equipment & Supplies; Toy & Hobby

Take Rate = referral fee per category(%) + {ad revenue($b)/3P GMV ($b)}

Market Share is Amazon's Retail Ecommerce sales share (US) by product category as % of total retail sales

Logistics Intensity = total logistics cost ($b)/3P GMV ($b)

Results/Insights

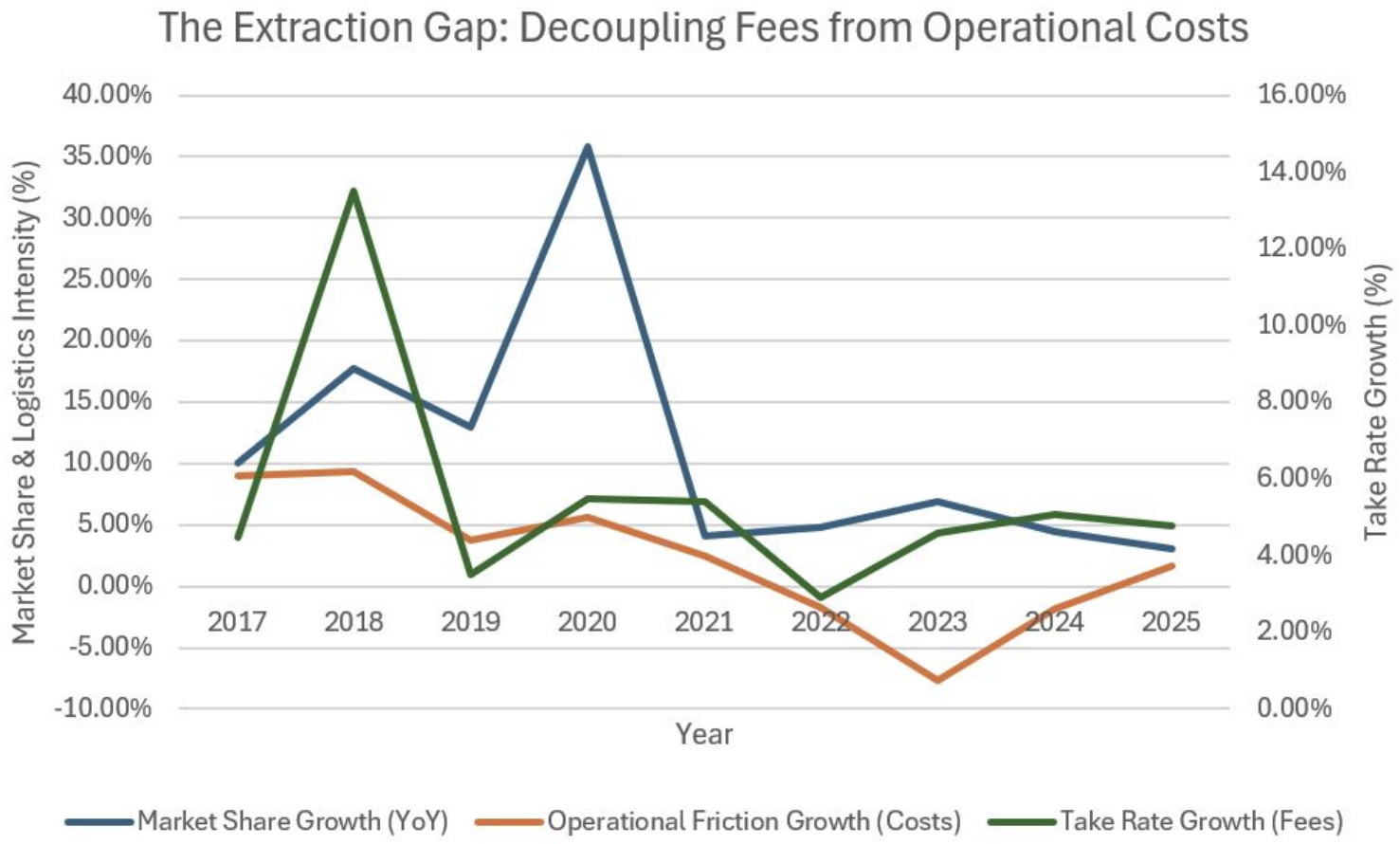
```
                          Estimate Std. Error t value   Pr(>|t|)
market_share              0.335134   0.049787 6.73139 1.6478e-09 ***
logistics_intensity_gmv 0.002175   0.000517 4.21097 6.1227e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.022311     Adj. R2: 0.66585
```

- The model identifies a clear decoupling of costs from pricing. While both variables are statistically significant, their economic impact is vastly different. Amazon isn't raising fees because they *must* cover rising costs; the impact of operational friction is essentially flat at 0.002. Instead, they are raising fees because they *can*. The market share coefficient of 0.335 proves that for every point of dominance Amazon gains, they extract significant additional rent from the 3P ecosystem.

- The change in take rate as a result of a 1% increase in market share is ~**154 times higher** than the change in take rate from a 1% increase in Operational Friction.

    - This is Amazon's **Gatekeeper Power** in action. Their size affords them the ability to block the path to the city (the customers) and impose a hefty toll (take rate) on all those who wish to pass.

Visual Representation of Regression



This graph showcases the percentage change in Market Share, Logistics Intensity, and Take Rate YOY. Because our variables exist at different magnitudes (Logistics Intensity averaging 65% while Market Share averages 12%) I decided to use a Year-over-Year Percentage Change to normalize the data. Its a dual axis graph with the percentage change in take rate on the right and the other 2 variables on the left.

The graph mainly depicts the aforementioned decoupling of costs and pricing. From 2021 onward, we observe an inverse relationship: as Amazon achieved economies of scale that reduced logistics intensity, the platform's 'Take Rate' continued its upward trajectory, tracking closely with market share dominance rather than marginal cost. This suggests that Amazon is capturing the entirety of its operational efficiency gains as **economic rent** rather than sharing them with the 3P ecosystem.

Their power as a **Gatekeeper** explains how they are able to raise fees even when logistics costs are flat or even negative.

Limitations Regarding Take Rate

Take Rate is intended to be the effective rate that Amazon charges 3P sellers as a fraction of the 3P gross merchandise value per category per year. However, these numbers are not public and almost certainly vary business to business. It is calculated for this project as a combination of the referral fee per category and a platform wide ad rate (assuming that ads are a necessary expense to show up in search results for customers) due to relatively scant data availability on an individual/firm level. One may very well need to reach out to an IID sample of businesses that use Amazon and request direct numbers from them to have a more exhaustive measure. That being said the aforementioned fees are very likely to be the largest ones that any 3P seller faces.

Constraints and Caveats

- Only 100 observations; 10 categories over 10 years (2016-2025)
- Adjusted R^2 is only .67 so the model only explains about 2/3 of the variation in take rate
- Take rate as defined for the regression is not an exact measure nor is it exhaustive
- Model assumes referral fees are very sticky
- Model assumes that primary vector of change for take rate beyond baseline category margins is ad revenue
- While the results are robust for the US e-commerce market, the coefficients may vary in international markets with different antitrust frameworks. Future research could apply this model to the EU and Asian regions to see if the 0.335 'Power Coefficient' holds globally
- The use of platform-wide logistics intensity serves as a high-level control for operational friction. While this does not capture category-specific shipping variances (e.g., heavy furniture vs. light electronics), this approach is intentionally conservative. It ensures that the model only attributes fee hikes to market share when those hikes exceed the platform-wide cost baseline
- Although Category Fixed Effects control for time-invariant differences, the model remains subject to potential omitted variable bias regarding time-varying factors, such as shifts in consumer preference or specific changes in Amazon’s Terms of Service for individual categories

Conclusions & The New Anitrust Reality

- **Evidence of Decoupling:** Empirical results confirm that fee increases are driven by market share, not operational costs. The 0.335 coefficient for dominance vs. the 0.002 for friction indicates that pricing has decoupled from service value.

- **The "Private Tax":** Because market share is a significantly stronger predictor of fees than logistics intensity, these charges function less like a service price and more like a **private tax** on e-commerce entry.

- **Algorithmic Power:** Control over demand allocation through algorithms determines seller success, creating a "winner-take-all" dynamic that reinforces Amazon's gatekeeper status.

- **Systemic Dependency:** High switching costs and a lack of viable outside options for sellers cement Amazon’s dominance, allowing for continued extraction of economic rents.

- **Antitrust Implications:** Traditional frameworks focused solely on consumer prices fail to capture the harmful extraction occurring on the "input side." In digital ecosystems, control over **access to a consolidated consumer base** is the new frontier of market power.

End

Ultimately, my research suggests that we are moving from an era of price-based monopolies to an era of access-based monopsonies. In this new landscape, the 'toll' for market entry is no longer determined by the cost of the road, but by the power of the gatekeeper.

Sources

10k filings
Emarketer
Statista